Tutorial Questions (Set-X)

Q1 (i) Let $\phi\colon \mathbb{R}^2 \to \mathbb{R}$ be given by $\phi(x,y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$. Show that $(D_1\phi)(x,y)$ and $(D_2\phi)(x,y)$ exist at every point $(x,y) \in \mathbb{R}^2$. However, show that ϕ is not continuous at $(0,0)$ and the total derivative $\phi'(0,0)$ does not exist.

(ii) Let $f(x,y) = \begin{cases} \frac{x^3}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$. Show that $D_1 f$ and $D_2 f$ are bounded functions on $\mathbb{R}^2$. Hence f is continuous. For an unit vector $u \in \mathbb{R}^2$ (with $\|u\| = 1$), show that the directional derivative $(D_u f)(0,0)$ exists. Further, show that f is not differentiable at $(0,0)$.

(iii) Let $m \in \mathbb{N}$ and $f(x) = \begin{cases} \frac{x^m y}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$. For $m = 1$, $f = \phi$ as in part (i). Determine the partial derivatives $(D_1 f)(x,y)$ and $(D_2 f)(x,y)$ at any point $(x,y) \in \mathbb{R}^2$. If $m = 2$, then show that f is continuous at $(0,0)$ and $(\nabla f)(0,0) = 0$, but the total derivative $f'(0,0)$ does not exist. For $m > 2$, show that $(D_1 f)(x,y)$ and $(D_2 f)(x,y)$ are continuous functions and $f'(0,0)$ exists.

(iv) Let $f\colon E \to \mathbb{R}^m$ be a mapping of a nonempty set $E \subseteq \mathbb{R}^n$ into $\mathbb{R}^m$ and

$$f(x) = \sum_{i=1}^{m} f_i(x) u_i \quad \text{for } x \in E.$$

If partial derivatives $D_j f_i(x)$ are bounded functions on E for $1 \leq i \leq m$ and $1 \leq j \leq n$, then show that f is continuous on E. Further, show that f is continuously differentiable on E if and only if partial derivatives $D_j f_i(x)$ exist and are continuous on E for $1 \leq i \leq m$ and $1 \leq j \leq n$.

(v) Let $E \subseteq \mathbb{R}^n$ be a nonempty open subset and $f\colon E \to \mathbb{R}$ be a differentiable function. Suppose f has a local maximum at $x_0 \in E$. Show that $f'(x_0) = 0$.

Q2 Let $f(x) = \begin{cases} x + x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$. Show that f is differentiable on $\mathbb{R}$ but not continuously differentiable at 0. Also, $f'(0) = 1$. However, show that f is not invertible in any neighbourhood of 0. (This shows that continuity of the total derivative is essential in the Inverse function theorem.)

Q3 (i) Let $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ given by $f(x_1,x_2) = (e^{x_1}\cos x_2, e^{x_1}\sin x_2)$ for $(x_1,x_2) \in \mathbb{R}^2$. Determine the Jacobian matrix $J_f(x_1,x_2)$ of f at (x_1,x_2). Show that f is locally invertible at every $(x_1,x_2) \in \mathbb{R}^2$. Further. determine an open set $E \subseteq \mathbb{R}^2$ such that f is one-to-one on E and $f(E) = \mathbb{R}^2 \setminus \{(x,0) : x \geq 0\}$. Does f globally invertible over E? Explain.

(ii) Let $f \colon \mathbb{R}^2 \to \mathbb{R}^2$ be given by $f(x_1, x_2) = (x_1^2 - x_2^2, 2x_1x_2)$ for $(x_1, x_2) \in \mathbb{R}^2$. Determine the Jacobian matrix $J_f(x_1, x_2)$ of f at (x_1, x_2). Further, show that $f \colon \mathbb{R}^2 \setminus \{(0,0)\} \to \mathbb{R}^2 \setminus \{(0,0)\}$ is locally invertible. However, f has no global inverse over $\mathbb{R}^2 \setminus \{(0,0)\}$. Let $E = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 > 0\}$. Then show that $f \colon E \to \mathbb{R}^2 \setminus \{(x_1, 0) : x_1 \geq 0\}$ is a C^1-diffeomorphism.

(iii) Let $f \colon \mathbb{R}^2 \setminus \{(0,0)\} \to \mathbb{R}^2 \setminus \{(0,0)\}$ be given by $f(x_1, x_2) = (\frac{x_1}{x_1^2+x_2^2}, \frac{x_2}{x_1^2+x_2^2})$ for $(x_1, x_2) \in \mathbb{R}^2$. Determine the Jacobian matrix $J_f(x_1, x_2)$ of f at (x_1, x_2). Show that f is invertible and find the inverse f^{-1} of f.

Q4 (i) Let $f(x,y) = f(x_1, x_2, y_1, y_2) = (x_1y_1^2 + x_2, x_1x_2^2 + y_1^2y_2^2)$. For $a = (-1,1)$ and $b = (1,1)$, we have $f(a,b) = 0$. Compute the total derivative $A = f'(a,b)$ and the Jacobian matrix $J_f(a,b)$. Show that A_y is invertible. Determine a function g from a neighbourhood of the point $a = (-1,1) \in \mathbb{R}^2$ into $\mathbb{R}^2$ such that $f(x, g(x)) = 0$. Further, determine $g'(a)$.

(ii) Show that the system of equations

$$\begin{aligned} 3x + y - z + u^2 &= 0, \\ x - y + 2z + u &= 0, \\ 2x + 2y - 3z + 2u &= 0, \end{aligned}$$

can be solved for x, y, u in terms of z. Determine $x = x(z), y = y(z)$ and $u = u(z)$. Can we express x, z, u in terms of y, or y, z, u in terms of x or x, y, z in terms of u? Explain.

(iii) Define $f \colon \mathbb{R}^3 \to \mathbb{R}$ by $f(x, y_1, y_2) = x^2y_1 + e^x + y_2$. Let $a = 0$ and $b = (1,-1)$. Show that $f(a,b) = f(0,1,-1) = 0$, $(D_1f)(a,b) \neq 0$ and that there exists a real-valued differentiable function g in some neighbourhood of $b = (1,-1) \in \mathbb{R}^2$ such that $g(b) = 0$ and $f(g(y_1,y_2), y_1, y_2) = 0$. Also, determine $(D_1g)(1,-1)$ and $(D_2g)(1,-1)$.